Exp No: 8	**Configuring user management and role-based access control (RBAC) in a Zoho Creator application.**

AIM:

To configure user management and role-based access control (RBAC) in a Zoho Creator application using permission sets, roles with role hierarchy, and data sharing rules.

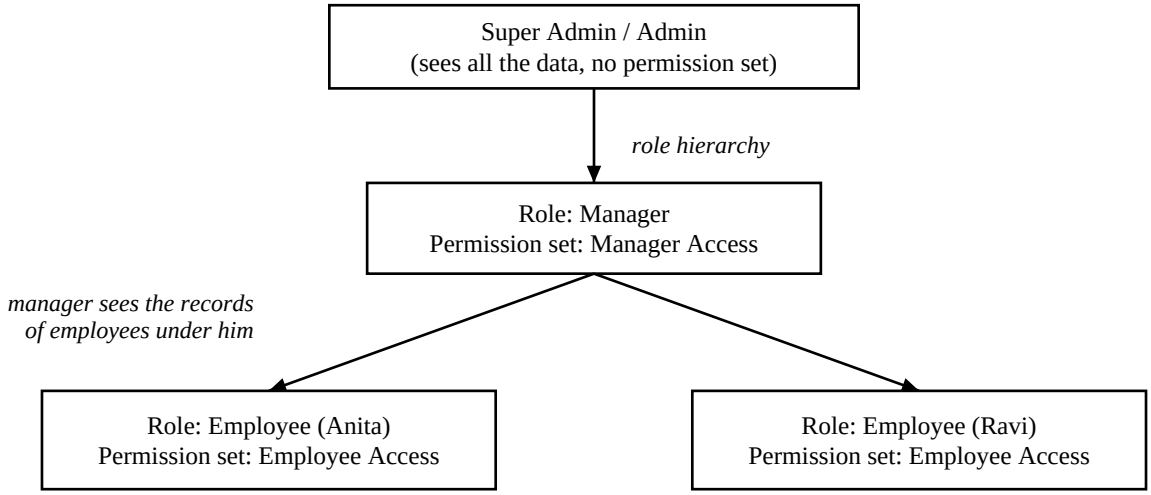
ALGORITHM / PROCEDURE:

Access control in Zoho Creator is based on three things: **Permissions** (what a user can do with the forms, reports, pages and fields), **Roles** (the position of the user in the organisation, arranged in a hierarchy) and **Data sharing** (which records of other users can be seen). A user is added to the application by the super admin / admin and is given a role and a permission set. Admins can see all the data and cannot be given permission sets. Custom permission sets and roles need a paid plan.

1. Create an application *Leave Manager* with a form *Leave Request* (fields: Employee Name, From Date, To Date, Reason, Status) and a report *All Leave Requests*.
2. **Plan the access:** an Employee can add a leave request and see only his own requests; he cannot change the Status field. A Manager can see the requests of the employees working under him and can change the Status (Approved / Rejected).
3. **Create permission sets:** open the application in edit mode → **Users & Control** → **Permissions** (Read and Write sets are available by default) → click **Add Permission**.
4. Create the set *Employee Access*: give a name and description, set the base access as Write (own data), enable the form *Leave Request* and its report, disable the other components, and under **Field Permission** make the field *Status* view only. Click **Save**.
5. Create the set *Manager Access*: enable all the components of the application and allow editing of the *Status* field. Keep API access as required. Click **Save**.
6. **Create roles:** go to **Roles** and add the role *Manager* (top level) and the role *Employee* under it. Enable the **role hierarchy** so that the manager can see the records of the employees under him.
7. **Data sharing rule (optional):** create a rule to share the *Leave Request* records of the Employee role with the HR role (for example, Read only).
8. **Add users:** go to **Users** → **Add Users**, enter the e-mail address of each person, select the role and the permission set (Manager → Manager Access, Employee → Employee Access) and send the invitation.
9. **Test:** log in to the live mode as an employee and as the manager (use a different browser or an incognito window). Check what each user can see and edit.
10. Note down the observations in the access table.

DIAGRAM:

Fig 1: Roles, role hierarchy and permission sets



Access table (expected result of the test)

Component / action	Employee Access	Manager Access
Add a Leave Request	Allowed	Allowed
View leave requests	Only his own records	Own records and records of employees under him
Edit the Status field	Not allowed (view only)	Allowed
Other forms / reports of the application	Hidden	Visible
Delete a record	Not allowed	Allowed

RESULT:

Thus user management and role-based access control were configured in the Zoho Creator application. Permission sets, roles with role hierarchy and data sharing rules were created, the users were added with a role and a permission set, and the access of each user was verified.